Forest Ethics & The Overstory

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Main Idea

01	**Introduction**	Plants, Plant Ethics, Forests, Forest Ethics
02	**Ecological Relationship**	Plant and Forest Ethics
03	**View Points**	Biocentrists, Anthropocentrists, and Sentientists
04	**Extinction of Species**	Causes, Replaceability of conscious and non-conscious creatures
05	**Renewable Resources**	Global Efforts to save forest, Impacts
06	**The Overstory**	Patricia Westerford's tree communication story

How Plants differ from Forests?

- Forest includes more than sum of their constituent plants and constituent living members
- Exercise influence that would not be the case if individual trees were situated in isolation
- Forests act as watersheds, absorbing rainfall, stimulants of rainfall, forms of precipitation
- Holds capability to modify climate of the lands
- Forests supply habitats for reproduction and growth to native species and others

What is Plant Ethics?

- Includes life processes of plants life-cycle (live, thrive and die) in large assemblages
- Consider benefits and the harms that befall plants and how humans shall treat them
- Few other main assemblages of trees forest:
  I. Bring plants to regular use,
  II. Leave plants alone or untreated
  III. Increase in number of befall plants relates to human act
- The conflicting views on whether forests are to be considered as resources and only resources are also part of plant ethics?

What is Forest Ethics?

- Consider all those areas of human activity that intersects with woods and forests including justification, rightful extent and character

Arguments:

- Attfield argues that "Not all plant ethics are part of forest ethics."
  I. Some conflicting issues such as the well-being of plants are relevant to both forest plants and non-forest plants. Eg. home gardens, fields, rivers, lakes, botanical gardens and parks.

- Attfield argues that "Not all forests ethics are part of plant ethics."
  I. Some branches of forest ethics concern the exploitation of forest products (eg. rubber and brazil nuts)
  II. Preservation of animal forest species

Relationship between Plant and Forest Ethics

- It would irrelevant to share that issues related to the ethics of forestry are not connected to plant ethics.
- Issues relating to cutting down and uprooting of trees that undermines the value of existing ones.
- These issues often need to be considered together if defensible overall judgments on issues such as deforestation and reforestation are brought up.

Biocentric stance in Ethics

- It considers trees has intrinsic value as it have good of their own.
- Attfield (biocentrist) argues that the trees and its interests shall be factored in decision-making.
- Biocentrist approach accounts for arising concern for individual trees and some in danger of being burned or chopped,
- Also considers ecological niches for future trees to grow and flourish
- Attfield approach to this concern that shall be grounded to all (even non-biocentrist)
- The permanent removal of forests is considered of higher concern rather than chopping down trees and to be readily replaced by newly grown.
- The replacement of trees by new ones are not always unethical.

Consequences of Deforestation

- Increase in CO_2 into the atmosphere as forest being cleared, reduces the ability to trap emissions ("carbon sink")
- Large proportion of trees with stored carbon are burned that releases CO_2
- There are very little prospect that CO_2 gets absorbed by growing vegetation at the same place.
- CO_2 levels rise from 280 ppm to ~ 400 ppm (parts per million) after industrial revolution
- IPCC warned increased levels of CO_2 has caused:

  I. Sea-level rise II. Global warming
  III. Flooding IV. Severe weather events
  V. Spread of diseases such as dengue fever, malaria, etc.
  VI. Extinction of species (flora and fauna) due to habitats destruction

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- Contributes to aesthetic and biodiversity losses
- Aesthetic loss accounts for the human love for trees, woods, and presence of forests not the value of relevant individual trees
- Aesthetic value is important to foster motivations for preserving forests.

Biocentrists, Anthropocentrists and Sentientists

- Biocentrists are concerned for human well-being (holds intrinsic value) and disruption of species that are often forced to migrate towards the poles until they ran out of habitat.
- In addition, they also shows concern for climate change
- According to them, trees are landmarks and carry symbolic value for humans and all flourishing trees holds intrinsic value.
- Anthropocentrists are concerned for humans and their values
- Sentientists are concerned for sentient animals (including humans) and their values
- Neither shows particular concern towards plants.
- Many threats are particularly to humanity and humans.

Extinction of Species

- Species extinction are greatly defined by human activities
- Species found on Coral reefs in rain-forests are at great risks.
  I. Deforestation/forest clearing
  II. Indiscriminate fishing
  III. Mining

- Imbalance in ecosystems cannot easily be reassembled
- Production of timber, paper and other products from chopping of forests can only be justifiable in a case when anthropocentrists and sentientist be condemned by biocentrists.
- The basis of moral acceptability of forestry relies onto the clear-felling of trees and ability of its regrowth, but in reality the prospect of regrowth reduced too severely

Replaceability of conscious and non-conscious creatures

- It is important to access the forestry practice analyzing the benefits and harms to humans, animals, plants including trees.
- In general, trees are considered replaceable but particular landmark trees holds irreplaceability status due to its historical or symbolic value
- Singer's argues that replaceability of conscious creatures (eg. cows) that are not self-conscious is not wrongful to kill them as long as it is done painlessly
- In order to compensate, raise and rear similar creatures
- The killing of self-conscious creatures is morally wrong
- Singer's view on trees applies a similar logic where trees are non-conscious and can be regarded as replaceable.

Renewable Resources

- Rearing for plants for fuel-wood is inexpensive to produce electricity.
- Planting Plantations for fuel-wood in itself a forestry practice
- Forest ethics shall not condemn this practice but broadly welcome with a view to satisfy human needs of poor people in developing countries
- Forest cannot be regarded as resources only as it is renewable resources with large proportion of species on the planets
- Renewable resources exists as long as sufficient forest is capable to renew itself through re-growth
- The ability of forest to renew itself in perpetuity making it possible for humans to benefit year after year. This corresponds to “sustainable management.”

Efforts towards saving forest

- Indonesia, Brunei, and Malaysia agreed upon to conserve forest of Borneo.
- Formation of ACTO (Amazon Cooperation Treaty Organization) to promote sustainable development in context of the UN Forum on Forest
- Susan proposal undertake important reforestation and biodiversity conservation works.
- This work would preserve forests, contribute to the absorption of greenhouse gases and reserve some of the endangered forest species.

Criticism and Arguments

- I agree to the Attfield arguments that all plant ethics are not part of forest ethics and all forest ethics are not part of plant ethics in entirety.
  I. Plants could be part of non-forestry and/or forestry resources.
  II. I'm convinced with the idea that Attfield portrayed about plants. It could be the part of forest whereas it could simply be one available at the botanical garden, parks, at the side of river, lakes, etc that isn't necessarily part of forest resources.
  III. Preservation of animals in the forest is not necessarily a part of plant ethics as to be more specific plants are typically assemblages of forest and non-forest plants.

- Although, there are difference between plant and forest ethics does not imply that they are not connected to each other.
  I. The uprooting or cutting down of trees does undermine both plant and forest ethics.

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- Biocentrist views does account for plants interest (intrinsic value), disruption of species (chopped trees, , climate change, and including ecological niches for future trees to grow and flourish.
- I agreed to the viewpoints from Biocentrist (Attfield) being conveyed to the audiences and readers.
- Attfield approach that these concerns shall be grounded to even non-biocentrist.
- The concern about clearing the forest is of higher concern than clear felling of trees and replantation on the same land.
- Although there are some concerns related to uprooting of trees that are quite old enough and has lower capability of replaceability.

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- Trees are considered “carbon sink” but replantation might not help in carbon captures as will take several years for some trees to be able to capture the carbon emissions as existing ones.
- Although, biocentrist think that “replacement of trees might not always be of ethical concern.” I believe this is only valid when the forests and its resources are used in a way that keeps ecosystem balanced.
- The consequences of deforestation has showed various burning issues. I believe that it is a valid concern.
- Anthropocentrists cares about the humans and their values while sentientists cares about sentient animals (including humans) and their values.
- The text says that neither shows values for plants where I do not completely agree with the text.

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- I believe that biocentrist views does align with caring for the plants because forest and its resources does include plants as well.
- Singer's compared the chopping of trees are replaceable as like compared to killing cows painlessly and as they both are non-conscious creatures
- I do not completely agree to Singer's viewpoint although they both are non-conscious creatures does not define that replacing them would cause the same level of harms to the environment.
- There weren't enough grounds for comparison between the two in the texts but I believe there are other ways to compare such as dry-mass, energy content, intrinsic value, potential, etc.

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- The use of renewable resources and its development has been a vital for to attain sustainability. Such as fuel-wood (part of forestry practice)
- The Global efforts toward sustainability practices by several NGOs, and UN bodies are playing key role along partnering with developing countries.
- These efforts have helped significantly to preserve forests and its species, and to reduce global emissions.

The Overstory

- The scientist Patricia Westerford's believes that trees have a way to communicate with one another
- She is rejected from the scientific community and becomes an outcast for believing in this claim
- They make Sexist claims such as "Girls doing science are like bears riding bikes. Possible, but Freakish."
- She sticks to what she believes and runs experiments consisting of taping numbered plastic bags over the ends of branches, then collecting them at measured intervals.

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- Other researchers finally catch up with her work and begin to replicate it and see that she is a genius and made a great discovery
- She then is accepted back into the scientific community and is praised and respected for her discovery

If the story of Patricia Westerford’s wasn't fiction how would it change environmental ethics?

- Perhaps, it would not make any difference
- Studies have shown that trees are connected to one another and are able to communicate in certain ways.
- They are able to warn one another from insects or disease , help share resources and nutrients with each other during harsh conditions
- The world already knows that trees are living giants that are capable of communicating with one another
- However mass deforestation is still a massive threat that they face currently

Souce: Do Trees Talk to Each Other?

The Overstory pt. 2

Analyzing objections a reasonable critic may come up with

Westerford's argument is compelling, but leaves important philosophical questions unanswered

Objection 1: Anthropocentrism

- **Anthropocentrism**- humans have a higher moral status than other organisms

- Objection- Even if forests are interconnected and shape ecosystems, humans still have a higher moral status, therefore, human interests and survival take priority over ecological preservation.

    - Critique- assumes environmental destruction is inevitable; Ecological life still have intrinsic value

Objection 2: Biocentrism

- Forest ecosystems function as interconnected wholes, not isolated organisms

- Objection- All living organisms have their own individual intrinsic value, so emphasizing the forest as a whole undermines individual organisms
    - Critique- Individual organisms inherent worth should not be diminished in order to acknowledge ecological wholes.

Objection 3: Instrumental Value

- Westerford explains that trees form interconnected systems that are conducive to living organisms beyond what we grasp.

- Objection- Trees sustain ecosystems and benefit other life, therefore their value is instrumental

    - Critique- Even if forests have instrumental value, they sustain other life and play a role in maintaining ecological balance

Objection 4: Moral Obligation

- Her research suggest that trees function together through underground networks, sharing nutrients
- Objection- Trees sharing nutrients in nature does not assign humans a moral obligation to protect them.
    - Critique- Understanding their function and interconnectedness reveals the consequences of destruction

Objection 5: No Individuals

- “There are no individuals.” - page 146
- Objection- Ecological interconnectedness undermines individual value by putting the system’s well-being ahead of the individuals that inhabit it.
    - Critique- protecting the ecological system is necessary for individual components to thrive within it.

Objection 6: Human Survival

- Forest systems sustain life and ecological balance
- Objection- It could be morally irrational to put ecosystems before immediate human needs in time of survival or shortage
    - Critique- Protecting ecological systems protects long term human survival

Thank You!